CS683 Programming Assignment II
Task 3: Integrating SPP with Prefetch-Aware Dynamic I-RRIP

1. Objective and implementation

Task 3 modifies the L2 replacement policy to recognize lines generated by L1D SPP, while retaining Dynamic I-RRIP's instruction-awareness. Its performance objective is to improve instructions per cycle (IPC) relative to the standalone components.

Prefetch-aware policy. Re-reference prediction values (RRPVs) range from 0 (strongly retained) to 3 (most eligible for eviction). In I-RRIP mode, instruction-demand fills enter at RRPV 1 and demand-data fills at RRPV 2. Instruction prefetches enter at RRPV 2; data prefetches enter at RRPV 3. Prefetch hits do not cause promotion, because a prefetch request does not itself establish demand reuse. Leader sets continue to use ordinary SRRIP, while follower sets dynamically select instruction-aware I-RRIP or SRRIP.

Dynamic selection. Every 32nd L2 set is a leader set. Demand-only leader statistics exclude prefetch, writeback, and translation requests. Every 4096 eligible accesses, the policy evaluates

$$P_I = \frac{\text{instruction accesses} + \text{instruction misses}}{\text{total accesses} + \text{total misses}}.$$

If $P_I > 0.25$, followers select I-RRIP; otherwise they select SRRIP. All observed evaluation epochs on the three measured traces selected I-RRIP.

2. IPC results

Benchmark	SPP only*	SPP disabled	Combined Task 3	Gain vs SPP
Trace 1	0.234213	0.232247	0.234111	−0.044%
Trace 2	0.253038	0.255522	0.257694	+1.840%
Trace 3	0.389045	0.382031	0.389107	+0.016%
Geometric-mean gain	–	–	–	+0.600%

*Standalone SPP IPCs use the Task 1 baseline measurements; the "SPP disabled" column uses the same L2 replacement policy without L1D SPP.

Interpretation. The combined configuration improves IPC relative to the SPP-disabled L2 policy on all three traces. Against standalone SPP, IPC improves on Traces 2 and 3 and is 0.044% lower on Trace 1. Its geometric-mean IPC gain relative to standalone SPP is approximately 0.60%.

3. Dynamic and prefetch-policy diagnostics

Statistic	Trace 1	Trace 2	Trace 3
Evaluation epochs	11	19	5
Epochs selecting I-RRIP	11	19	5
Mode switches	0	0	0
L2 prefetch fills (policy counter)	90,711	52,359	72,975
Prefetch hits not promoted	26,976	7,179	4,623

Nonzero fill and unpromoted-hit counters confirm execution of the prefetch-aware paths. Zero observed mode switches mean the 25% threshold was not crossed during these measurement windows, not that switching would be unnecessary on other workloads.

4. Performance visualization

Figure 1 compares absolute IPC across three configurations. Figure 2 shows load miss intensity at L1D and L2C for the combined configuration; these metrics characterize different levels of the cache hierarchy and should not be directly summed.

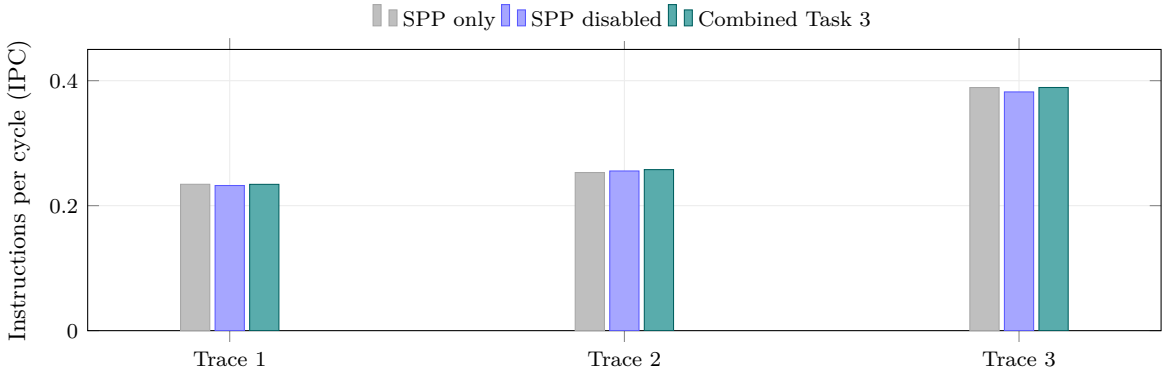


Figure 1: IPC across the three benchmark traces at the prescribed 25M/25M instruction settings.

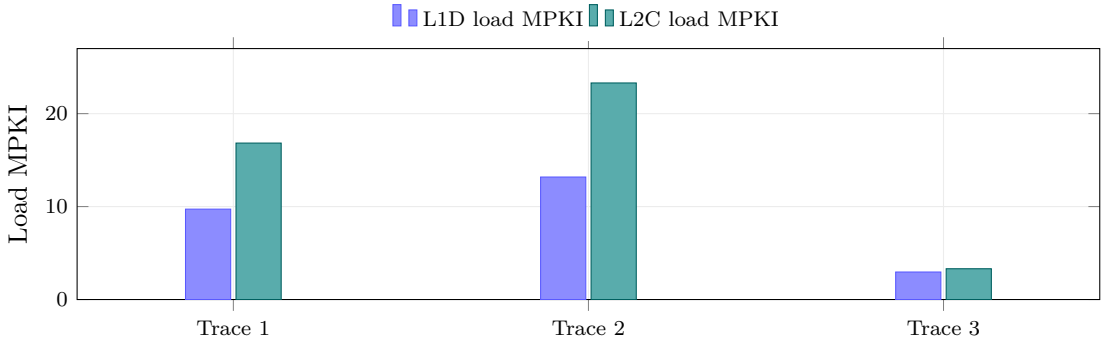


Figure 2: Load misses per thousand instructions under the combined policy.

5. Cache and prefetch metrics

MPKI is the number of cache misses per thousand measured instructions. The following measurements are from the three completed Task 3 simulation logs.

Metric	Trace 1	Trace 2	Trace 3
L1D load misses	243,122	329,502	74,004
L1D load MPKI	9.72488	13.18010	2.96016
L2C load MPKI	16.83290	23.30040	3.31368
L2C instruction load MPKI	10.08180	14.79200	1.07980
L2C data load MPKI	6.75108	8.50840	2.23388
L1D SPP accuracy (%)	56.2569	48.8721	45.6523
L1D useful prefetches	32,094	11,353	12,002
L1D prefetches issued to lower level	57,049	23,230	26,290

Instruction and data L2C load MPKIs sum to total L2C load MPKI, subject to rounding.

L1D prefetch coverage. We use

$$\text{Coverage} = \frac{\text{L1D load misses}_{\text{baseline}} - \text{L1D load misses}_{\text{combined}}}{\text{L1D load misses}_{\text{baseline}}} \times 100\%.$$

The baseline measurements provide rounded MPKIs (10.678, 13.538, and 3.403). Consequently, the following percentages are *approximations* computed using MPKI rather than exact baseline miss counts.

Trace	Baseline L1D MPKI*	Combined L1D MPKI	Approx. coverage
Trace 1	10.678	9.72488	8.93%
Trace 2	13.538	13.18010	2.64%
Trace 3	3.403	2.96016	13.01%

*Baseline MPKIs are drawn from the Task 1 baseline measurements under the same specified instruction counts. Exact coverage requires the baseline's unrounded L1D load miss counts.

Accuracy definition. L1D SPP accuracy is the simulator's useful load prefetches divided by prefetch requests issued to the lower level. This differs from the L2 prefetch accuracy metric, which measures demand usefulness at another cache level.

6. Build and reproducibility

The implementation consists of `prefetcher/spp.l1d_pref`, `replacement/dyn_irrip.l2c_repl`, and modified `inc/cache.h`, `src/cache.cc`, and `src/main.cc`. Simulator modifications propagate instruction-line metadata into the replacement policy and support extensionless raw traces. These common simulator modifications are required to reproduce the combined configuration.

Build: `./build_champsim.sh spp dyn_irrip task3`

Run: `-warmup_instructions 25000000 -simulation_instructions 25000000 -traces traces/TraceN`

Validation status. Compilation and three full-length trace runs completed with exit code 0. Coverage is approximate because only rounded baseline load MPKIs are available. Trace inputs are not embedded in the source distribution.